Chapter 01

Introduction & Linux Foundations

Episode 1.01

Episode title: **What is Linux**

Objective: 1.1

What is an OS?

An **operating system (OS)** is a **program** that runs on a computer and provides a **software platform** on which **other programs** can run.

-NIST SP 800-86

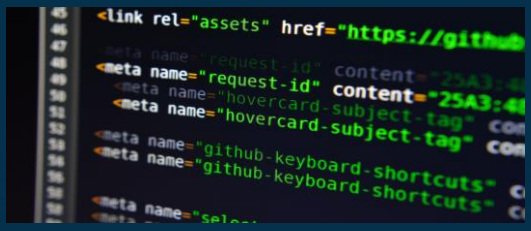
Linux Origins

1990s

2000s

2010s

2020s



Linux Uniqueness

Open source

Multiple distributions

Large community

Wide enterprise use (servers, containers, cloud instances)

Episode 1.02

Episode title: **Linux Distributions**

Objective: 1.1

Distributions

Combination of the **Linux kernel** with **system libraries**, **utilities**, **applications**, and **package management**

Ready to deploy

Distributed through dedicated websites and repositories

Kernel

Core of the operating system

Interface between hardware and software

Manages memory, processes, devices, system calls, and security

Monolithic design running in privileged kernel space

Modular support via loadable kernel modules (runtime add/remove)

System Libraries

Pre-compiled, reusable code

Interface between applications and the kernel

Provide access to system resources

- Memory, storage, devices

Standardized APIs used by applications

Utilities

Command-line tools for specific system tasks

Used for system maintenance and management

Perform common operations:

- File access, copying, and text search

Applications

- End-user software that performs specific tasks
- Run on top of system libraries and the kernel
- Provide direct user interaction
- Examples include web browsers, media players, text editors, and office suites

Package Managers

Automate software installation, removal, and updates

Manage dependencies and package configuration

Verify software authenticity via trusted repositories

Examples include APT, YUM/DNF, PACMAN, and PORTAGE

Popular Distributions

Ubuntu

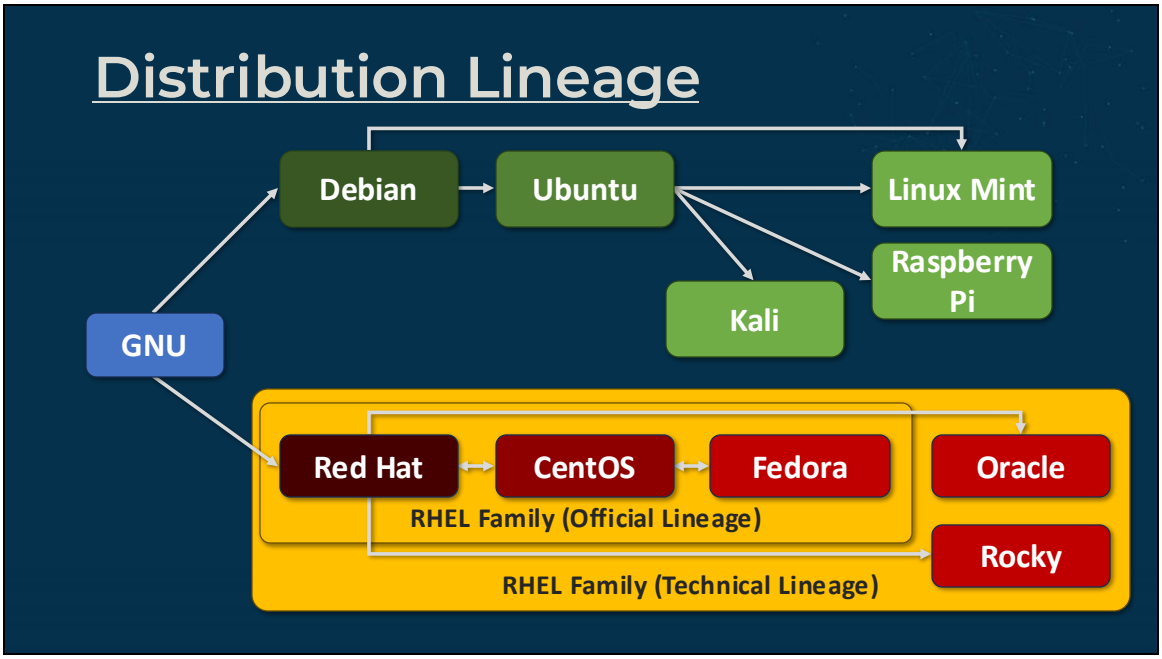
Debian

Linux Mint

Fedora

Arch Linux

Red Hat Enterprise Linux


Distribution Lineage
Debian
Ubuntu
Linux Mint
Raspberry Pi
Kali
GNU
Red Hat
CentOS
Fedora
Oracle
RHEL Family (Official Lineage)
Rocky
RHEL Family (Technical Lineage)

Episode 1.03

Episode title: **Open Source & Licensing**

Objective: 1.1

Important Terminology

GNU Public License

GPLv1 - 1989

GPLv2 - 1991

GPLv3 - 2007

Closed on Open

Linux is open source

Software may not be

Understand the licensing

Free = Open Source

Proprietary software

Select and implement carefully

Episode 1.04

Episode title: **Graphical vs. CLI Environments**

Objective: 1.1

Graphical User Interface (GUI)

Point-and-click-based

Graphical representation

Familiar to many users

Less efficient for repetitive work

Limited depth of control

GUI Components

Display managers

Window managers

X Server

Wayland

GNOME and KDE

Command Line Interface (CLI)

Command-based

Text representation

Less familiar to many users

More efficient for repetitive work

Deep control

Episode 1.05

Episode title: **Linux Boot Process**

Objective: 1.1

Linux Boot Process

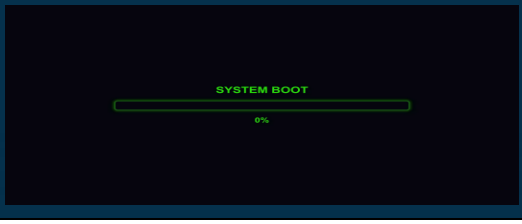
BIOS/UEFI

- Local disk vs. PXE

Bootloader

Kernel

Initial RAM disk (initrd)



Kernel Parameters

- root=
- ro
- quiet
- debug
- splash
- single
- init=/bin/bash

Kernel Parameters

root=/dev/sda1 quiet ro

initrd

initramfs

systemd

init

Episode 1.06

Episode title: **Filesystem Hierarchy Standard**

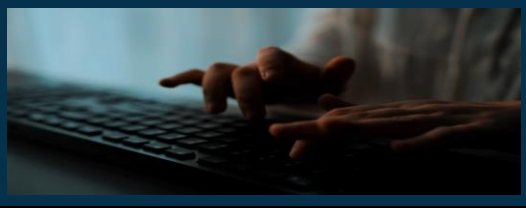
Objective: 1.1

File System Hierarchy

Linux directory structure roadmap

Defines where specific files or file *should* be

Used by nearly all Linux distributions



FHS Structure

/	/lib
/bin	/proc
/boot	/sbin
/dev	/tmp
/etc	/usr
/home	/var

Episode 1.07

Episode title: **Installing Linux**

Objective: 1.1

DEMO

Demonstrate installing Ubuntu in a VM

Discuss server architectures

Episode 1.08

Episode title: **Updating Linux**

Objective: 1.1

Demo

Demonstrate updating linux with APT

Discuss other update engines

Episode 1.09

Episode title: **Customizing Linux Part 1**

Objective: 1.1

Post-Installation Customizations

Why post-install configuration matters

Common customization categories

Security, stability, and manageability goals

IUR

Installation, Update, and Removal (IUR) of software is essential

- Repositories/ Source (code)
- Package managers
- Package dependencies and conflicts
- Language-specific managers

Repository Mgmt.

Repository management

Package and repository exclusions

Update alternatives

Traditional vs. Sandbox

Software configuration the traditional way (package managers)

Software configuration in isolation (sandboxes)

Episode 1.10

Episode title: **Customizing Linux Part 2**

Objective: 1.1

User Account & Privilege Management

Create users (adduser), manage groups (usermod, groupadd)

Sudoers and least privilege

SSH keys vs passwords

SSH Hardening

Install OpenSSH server

Disable root login, disable Password Authentication

Use Fail2Ban or UFW rate limiting

Network & Hostname

Set hostname and update /etc/hosts

Configure static IP or DHCP via netplan (Ubuntu)

Verify with ip addr and systemd-resolve --status

Time, Locale & NTP

Set timezone (timedatectl)

Enable NTP/time sync (systemd-timesyncd or chrony)

Configure locales

Firewall & Host Security

Enable and configure UFW or iptables/nftables

AppArmor or SELinux status

Install auditd for logging

Storage & Filesystems

Configure LVM logical volumes and resize if needed

Create swap or swapfile, mount points in /etc/fstab

Mount and persist additional disks

Guest Tools, Backups & Monitoring

Install VirtualBox Guest Additions or cloud agents

Configure backups (rsync, borg, snapshots)

Set up monitoring (Prometheus, cloud agents)

Final Checklist & Exam Tips

Confirm updates, user accounts, and SSH keys

Verify firewall and AppArmor, ensure time sync

Validate mounts, backups, and monitoring

Episode 1.11

Episode title: **Customizing Linux Demo**

Objective: 1.1